

Generation Interconnection System Impact Study Report

For Queue Project Q23-2276

West Asheville 100 kV

February 18, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2276. The Interconnection Customer proposes a 186.1 MW battery energy storage facility interconnecting at the West Asheville 100 kV substation in the Indiana area. The request was received on January 13, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 687 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,213,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W ASHEVILLE – WEBSTERVILLE SUBSTATION 100 kV	MONTPLIER ELECTRIC GENERATING STA SUBSTATION 230 kV/1 kV Autotransformer	135.2	125	25.9
ANCHOR HOCKING – WEBSTERVILLE SUBSTATION 100 kV	POWERLINE ELKORN – WEBSTEVILLE 100 kV	125.5	226	19.0
CHINA – BEDFORD RURAL 100 kV	MONTPELIER ELECTRIC GENERATING STA 230/1 kV XFMR	123.1	300	19.5
HARTFORD CITY – BEDFORD RURAL 100 kV	MONTPELIER ELECTRIC GENERATING STA 230/1 kV XFMR	122.3	251	14.2
WEST MILTON – WEST RIVERSIDE 100 kV	FITCHBURG – WEST RIVERSIDE 100 kV	121.0	139	8.1
DEER CRK – PONNEQUIN 100 kV Line	MONTPELIER ELECTRIC GENERATING STA 230-1 kV XFMR	119.3	276	19.5
COSO 7 – WEST ASHEVILLE 100 kV line	MONTPELIER ELECTRIC GENERATING STATION 230-1 kV XFMR	118.7	257	19.6
CHINA – PONNEQUIN 100 kV	MONTPELIER ELECTRIC GENERATING STATION 230 kV/1 kV Auto	117.4	301	19.5
COSO 7 – MONTPELIER ELECTRIC GENERATING STA 100 kV Line	MONTPELIER ELECTRIC GENERATING STATION 230/1 kV Auto	116.5	284	19.6
OAK PLAINS – EASTMONT 100 kV L	MONTPELIER ELECTRIC GENERATING STA SUBSTATION – MUSKOGEE 500 kV	113.5	170	6.5
DUTTON PUMP – BOURNE 100 kV	BUTLER MUNICIPAL POWER PLANT – SAVONA 100 kV	113.3	266	11.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
KLAMATH CO-GEN – BOURNE 100 kV	SAVONA SUBSTATION – BUTLER MUNICIPAL POWER PLANT 100 kV	111.9	282	11.9
SHERMAN ST – KLAMATH CO-GEN 100 kV	BUTLER MUNICIPAL POWER PLANT – SAVONA 100 kV Line	111.0	295	11.9
SHERAN ST – EASTMONT 100 kV	BUTLER MUNICIPAL POWER PLANT – SAVONA 100 kV line	110.3	306	11.9
DUTTON PUMP SUBSTATION – LEYENDECKER 100 kV line	E DRAINAGE CANAL – ST. LAWRENCE 230 kV	107.4	356	7.3
N TIFTON – W FORD FLAT 100 kV	MONTPELIER ELECTRIC GENERATING STATION 230/1 kV Autotransformer	106.3	146	26.5
TIMBER SWITCH – W MELROSE 230 kV	SWAN VLY – TREGO 500 kV	101.4	929	5.9
MUSKOGEE – TIMBER SWITCH 230 kV	SWAN VALLEY SUBSTATION – TREGO 500 kV	101.3	891	6.0
MUSKOGEE 500 kV/230 kV XFMR	SWN VALLEY – MUSKOGEE 500 kV	100.0	1,703	10.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W ASHEVILLE – WEBSTERVILLE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$37,839,000	19% (\$7,342,000)
ANCHOR HOCKING – WEBSTERVILLE 100 kV L	Rebuild line with higher-capacity conductor	\$6,520,000	16% (\$1,059,000)
CHINA – BEDFORD RURAL 100 kV	Rebuild line with higher-capacity conductor	\$4,938,000	9% (\$428,000)
HARTFORD CITY – BEDFORD RURAL 100 kV	Rebuild line with higher-capacity conductor	\$4,767,000	9% (\$416,000)
W RIVERSIDE – W MILTON 100 kV Line	Rebuild line with higher-capacity conductor	\$1,584,000	14% (\$227,000)
DEER CRK – PONNEQUIN 100 kV Line	Rebuild line with higher-capacity conductor	\$3,895,000	9% (\$338,000)
COSO 7 – WEST ASHEVILLE 100 kV line	Rebuild line with higher-capacity conductor	\$9,586,000	10% (\$969,000)
CHINA – PONNEQUIN 100 kV	Rebuild line with higher-capacity conductor	\$3,341,000	9% (\$290,000)
COSO 7 – MONTPELIER ELECTRIC GENERATING STA 100 kV Line	Rebuild line with higher-capacity conductor	\$20,044,000	10% (\$2,026,000)
OAK PLAINS – EASTMONT 100 kV L	Rebuild line with higher-capacity conductor	\$23,884,000	12% (\$2,789,000)
DUTTON PUMP – BOURNE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$2,703,000	14% (\$382,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KLAMATH CO-GEN – BOURNE 100 kV	Rebuild line with higher-capacity conductor	\$5,816,000	14% (\$821,000)
SHERMAN ST – KLAMATH CO-GEN 100 kV	Rebuild line with higher-capacity conductor	\$3,558,000	14% (\$503,000)
SHEMAN ST – EASTMONT 100 kV	Rebuild line with higher-capacity conductor	\$3,371,000	14% (\$476,000)
DUTTON PUMP – LEYENDECKER 100 kV	Reconductor line	\$2,367,000	5% (\$115,000)
N TIFTON – W FORD FLAT 100 kV	Reconductor line	\$4,300,000	22% (\$958,000)
WEST MELROSE – TIMBER SWITCH SUBSTATION 230 kV line	Reconductor line	\$36,634,000	3% (\$981,000)
MUSKOGEE SUBSTATION – TIMBER SWITCH 230 kV	Reconductor line	\$8,135,000	3% (\$217,000)
MUSKOGEE 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$34,916,000	2% (\$876,000)
	Total allocated to project		\$21,213,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.